

CivicConnect Project Engineering Document (PED) v1.0 — Engineering Baseline

SOFTWARE ENGINEERING 381 — SEN381
CIVICCONNECT
Project Engineering Document (PED)
Version 1.0 — Engineering Baseline

Jordan Naidoo (602463), Darrin Everette
Mokuena(601172)

Contents

Problem and Business Need	2
Problem Statement.....	2
Business Need	2
Project Business Objective.....	3
Stakeholder Analysis.....	3
Stakeholder Register	3
STK-01 — Requesters.....	4
STK-02 — Service Staff	4
STK-03 — Management / Oversight.....	5
STK-04 — System / Operational Administrator.....	5
Stakeholder Conflicts and Competing Expectations.....	6
Scope Baseline	6
Purpose of the Scope Baseline.....	6
In-Scope Business Capabilities	6
Supporting Scope.....	7
Out-of-Scope	7
Deferred / Future Scope.....	8
Deliberate Scope Exclusion — Direct WhatsApp Integration	8
Project Constraints	9
Purpose of Constraint Analysis.....	9
Constraint Register	9
Constraint Interactions and Trade-Offs.....	9
Initial Risk Register.....	10
Risk Management Approach.....	10
Risk Rating Method.....	11
Initial CivicConnect Risk Register.....	11
Highest-Priority Risk Analysis.....	14
Functional Requirements.....	15
Non-Functional Requirements.....	15
Acceptance Criteria.....	16
Initial RTM.....	16
Engineering Decision Log.....	17
Decision Status	17
Initial Engineering Decision Register.....	17
Forward Engineering Considerations.....	17
PED v1.0 Baseline Review and Sign-Off	18

Problem and Business Need

Problem Statement

The organization currently manages community service requests through disparate communication channels, including email, telephone, WhatsApp messages, spreadsheets and paper-based records. This approach makes it difficult to keep a consistent record of each request during its lifecycle. As a result, requests may be duplicated, missed, misplaced or wrongly handled, and requesters have limited insight into whether their requests have been received, assigned, delayed, resolved or closed.

The central problem is therefore not the lack of software, but the lack of a controlled, traceable and consistent process for managing service requests. Staff need clearer ownership and prioritization of their work, requesters need greater visibility and feedback, and management needs reliable information about outstanding, overdue and completed requests. The current fragmented process also creates accountability challenges because status changes and actions are not consistently captured in a controlled environment.

CivicConnect will solve this problem by providing a controlled digital platform through which service requests can be submitted, managed, monitored and reported. The goal is to improve visibility and accountability without imposing an excessive technical, operational or financial burden on the organization.

Business Need

The organization needs a single system that will embrace the entire service request lifecycle and replace the reliance on disparate manual methods of communication and record keeping. The system to be selected must enable requesters to track their requests, allow authorized staff to manage and update requests in a controlled way and provide management with the information they need to monitor service activity and accountability.

The intended business value of CivicConnect is:

Business need	Intended value
Centralized request management	Creates one controlled source of service-request information.
Improved requester visibility	Allows requesters to see the current status and history of their requests.
Clear staff responsibility	Makes it easier to identify who is responsible for a request.
Controlled status management	Reduces inconsistent or untraceable status changes.
Improved accountability	Maintains information about actions, comments and resolution activity.
Better management oversight	Provides information on open, overdue, resolved and closed requests.
More reliable reporting	Reduces dependence on manually assembled and inconsistent reports.

These value areas are directly related to the weaknesses identified in the existing process and to the minimum requester, staff and management capabilities required for CivicConnect.

Project Business Objective

The business goal of CivicConnect is to design a controlled and traceable service request management platform through which requests are made, assigned, monitored, updated and closed whilst providing appropriate visibility to requesters, staff and management.

The project can only be considered as valuable, if it meets the needs of the stakeholders within the agreed confines of scope, schedule, cost/resource, quality, security and operational constraints. The Master Project Brief explicitly refers to the fact that project success is not merely defined as whether the final application runs or appears visually complete.

Stakeholder Analysis

Stakeholder analysis is required in M1 because the requirements baseline must derive from the actual needs of the stakeholders instead of basing it simply on assumptions made by the development team. The Milestone 1 brief specifically requires stakeholders, their needs, influence/interest and meaningful conflicting expectations to be identified.

The Systems Engineering Body of Knowledge explains that stakeholder needs include: stakeholder expectations, wants, perceived constraints and desired system behavior. These needs should subsequently be transformed into more structured and formal requirements (SEBoK, 2025; ISO/IEC/IEEE, 2018).

Stakeholder Register

ID	Stakeholder	Interest	Influence	Key Needs / Expectations
STK-01	Requester	High	Medium	Simple request submission, appropriate categorization, request status visibility, history and meaningful feedback.
STK-02	Service Staff	High	High	Access relevant requests, locate requests efficiently, view details, accept/assign responsibility, update statuses and record work performed.
STK-03	Management / Oversight	High	High	Reliable information about service activity, outstanding and overdue work, completed requests and organizational accountability.
STK-04	System / Operational Administrator	Medium	High	Controlled access, maintainability, configuration control, reliable operation, logging and eventual recovery capability.

STK-01 — Requesters

Requesters are the people who report matters concerning the provision of services to the organisation. The primary interest of requesters is having a reliable and understandable method of submitting a request and understanding what happens after the request is submitted.

The requester should be able to:

- submit an appropriate service request;
- categorise the request;
- view its current status;
- view previously submitted requests; and
- receive meaningful feedback when a request is accepted, rejected, updated or completed.

These expectations are directly established by the Master Project Brief

Main need:

Visibility and confidence that the request has entered a controlled process.

Requirement implications:

The requirements should therefore include functions for request submission, categorisation, status viewing, request history and appropriate feedback.

STK-02 — Service Staff

Service staff require access to requests relevant to their areas of responsibility and sufficient information to understand and perform on each request.

The Master Project Brief requires staff to be able to search, filter or sort requests, view full details of requests, take ownership, update requests by making controlled status changes, log actions or comments and resolve or close requests if authorized

Main need:

An efficient but controlled way of managing assigned service work.

Requirement implications

Requirements should cover:

- authorised request access;
- request searching/filtering/sorting;
- responsibility or assignment;
- controlled status transitions;
- comments/actions;
- resolution information; and
- authorised closure.

STK-03 — Management / Oversight

Management needs reliable information on service activity, rather than having to create reports manually from a number of different sources.

The required management capabilities are identifying open, overdue, resolved and closed requests, viewing service activity by useful dimensions such as category and status, and getting sufficient information to enable accountability and service-performance analysis

Main need:

Reliable oversight and accountability information.

Requirement implications:

Later requirements should provide management with appropriate summary and reporting capabilities without automatically granting unrestricted access to every piece of service-request information.

STK-04 — System / Operational Administrator

The system will eventually require operations such as maintaining configuration, controlling access and supporting deployment and operation.

Security practice also requires that the access privileges of a user or process are restricted to what they need to perform their task. The OWASP principle of least privilege recommends that permissions are restricted to those necessary to perform the required task, rather than providing excessive access (OWASP Foundation, n.d.).

Main need:

Maintainability and controlled operation of CivicConnect.

Requirement implications:

The M1 requirements should recognise the need for controlled authentication/authorisation and secure information handling. However, the exact authentication mechanism, access control implementation and technology should be decided in a later milestone.

Stakeholder Conflicts and Competing Expectations

Stakeholders do not necessarily want exactly the same things. Requirements engineering must therefore identify areas where satisfying one expectation could negatively affect another.

Conflict	Competing Expectations	Engineering Effect
Visibility vs confidentiality	Requesters and management want useful information, while some request information may be sensitive.	Later access rules must balance useful visibility with restricted access.
Ease of use vs accountability	Staff need fast status updates, but management needs traceable changes.	Updating requests should remain convenient while retaining meaningful history.
More functionality vs project capacity	Stakeholders may benefit from extra features, while the project has limited time and resources.	Minimum business capabilities should be prioritized and non-essential features controlled or deferred.
Fast resolution vs realistic workload	Requesters may want immediate outcomes, while staff may need to priorities multiple requests.	CivicConnect should provide meaningful status visibility without making unrealistic service promises.
Detailed oversight vs information exposure	Management requires oversight, but access to sensitive details should be appropriately limited.	Reporting and access requirements should distinguish necessary management information from unnecessary sensitive detail.

Identifying these conflicts now is important because stakeholder needs become inputs to later requirements and design decisions rather than being treated as independent wish lists (SEBoK, 2025).

Scope Baseline

Purpose of the Scope Baseline

The scope baseline defines what we are currently delivering and what is being specifically excluded or deferred. A clear scope statement is also important in professional practice for new requirements to provide the basis for assessment of whether they are within the scope of the project. Without clear project boundaries, teams may struggle to assess whether newly identified functionality within the project falls inside or outside the approved effort, increasing the possibility of uncontrolled scope expansion (Burek, 2008).

In-Scope Business Capabilities

The following functionality forms the proposed committed scope

Requester Scope

SC-IN-01 — Submit a service request with the required information.

SC-IN-02 — Categorise a service request using a controlled category mechanism.

SC-IN-03 — View the current status of submitted requests.

SC-IN-04 — View a history/list of previously submitted requests.

SC-IN-05 — Receive meaningful feedback when a request is accepted, rejected, updated or completed.

Staff Scope

SC-IN-06 — View service requests relevant to authorised staff.

SC-IN-07 — Search, filter and/or sort requests using useful criteria.

SC-IN-08 — View complete request details.

SC-IN-09 — Assign or accept responsibility for requests.

SC-IN-10 — Update requests through controlled status transitions.

SC-IN-11 — Record relevant actions, comments and resolution information.

SC-IN-12 — Resolve or close requests where authorised.

Management / Oversight Scope

SC-IN-13 — View useful service-activity information.

SC-IN-14 — Identify open, overdue, resolved and closed requests.

SC-IN-15 — View information according to useful dimensions such as request category and status.

SC-IN-16 — Access sufficient information to support accountability and service-performance analysis.

These sixteen capabilities reflect the minimum CivicConnect business capabilities established in the Master Project Brief.

Supporting Scope

Certain supporting capabilities will also be required to provide controlled and usable versions of the above functionality.

SC-IN-17 — Authorised users must be distinguishable according to their permitted responsibilities.

SC-IN-18 — Service-request information and relevant lifecycle history must persist reliably.

SC-IN-19 — Significant request status and responsibility changes must be traceable.

SC-IN-20 — Required request information must be appropriately validated.

SC-IN-21 — Sensitive information and credentials must be handled through appropriate security controls.

These supporting items are consistent with the Master Brief's requirements for traceability, security and lifecycle accountability.

They do not specify the final architecture, programming language, database, technology to authenticate or deploy. ISO/IEC/IEEE requirements-engineering guidance differentiates between defining and managing requirements and prematurely fixing a particular implementation solution (ISO/IEC/IEEE, 2018).

Out-of-Scope

The following items are proposed deliberate exclusions from the initial CivicConnect baseline:

SC-OUT-01 — Native Android application.

SC-OUT-02 — Native iOS application.

SC-OUT-03 — Automatic request creation directly from WhatsApp.

SC-OUT-04 — Automatic service-request creation from social-media platforms.

SC-OUT-05 — SMS-based request submission and management.

SC-OUT-06 — IoT/sensor-generated service requests.

SC-OUT-07 — Full procurement and financial-management functionality.

SC-OUT-08 — Full stock/inventory-management functionality.

SC-OUT-09 — Multi-organisation commercial SaaS functionality.

SC-OUT-10 — Replacement of unrelated organisational systems.

Every additional feature raises additional requirements, design, security, implementation, testing, documentation, deployment and maintenance obligations. The Master Brief specifically points out that a smaller controlled solution can have more Software Engineering than an unfinished solution that has too much functionality.

Deferred / Future Scope

The following capabilities have potential stakeholder value but are **not part of the current committed baseline**:

SC-FUT-01 — Native mobile applications.

SC-FUT-02 — WhatsApp or external communication-channel integration.

SC-FUT-03 — SMS notifications.

SC-FUT-04 — Geographic/map-based request visualisation.

SC-FUT-05 — Advanced analytical dashboards.

SC-FUT-06 — More advanced escalation and workflow automation.

SC-FUT-07 — Integration with external asset-management or maintenance systems.

SC-FUT-08 — Advanced media/attachment functionality.

Deliberate Scope Exclusion — Direct WhatsApp Integration

One deliberate proposed exclusion is direct integration between CivicConnect and WhatsApp.

Whilst it is one of the current contributing factors to the fragmented request-management environment, the business problem does not dictate that CivicConnect must integrate with every available communication channel. The priority task is instead to establish a controlled, traceable central request-management process.

Direct integration with an external platform would add interface dependencies, testing requirements, security considerations, operational dependencies and potential cost or vendor constraints. Adding functionality without control over its impact on project scope exposes schedule and other objectives to risk (Abramovici, 2000).

Direct WhatsApp integration is therefore proposed as future scope, making it possible to prioritise the establishment of a core service request lifecycle before introducing changes that impact it. The decision can be revisited through controlled change if sufficient stakeholder value justifies impact.

Project Constraints

Purpose of Constraint Analysis

Constraints define conditions within which the system must be engineered. This specifically requires scope, schedule, cost/resources, quality and security constraints to be recorded along with their implications for engineering. Constraints are significant because they affect future architecture, technology, implementation, testing and deployment decisions.

Constraint Register

ID	Constraint	CivicConnect Constraint	Engineering Implication
CON-01	Scope	Approved functionality must remain within the baselined project boundaries.	New functionality must not be added informally after baseline.
CON-02	Schedule	CivicConnect must progress through the four formal SEN381 milestones within the academic delivery period.	Work must be prioritized and uncontrolled expansion avoided.
CON-03	Cost	Free or low-cost development/deployment options should be preferred where practical.	Later technology decisions must investigate limits and likely operational cost.
CON-04	Quality	Quality attributes must eventually be demonstrated using measurable evidence.	NFRs and acceptance criteria must be measurable and later linked to tests.
CON-05	Security	Security must be considered throughout the lifecycle rather than added immediately before deployment.	Access, sensitive information, secrets, dependencies and security verification need consideration from early requirements onward.
CON-06	Technology	No final stack or architecture is prescribed.	Technology must be selected later using requirements, evidence and trade-off analysis.
CON-07	Institutional Environment	The institution does not guarantee support or compatibility for every technology or external platform selected by a team.	Compatibility must be investigated before final technology commitment.
CON-08	Configuration Management	Controlled project changes must be represented progressively through repository evidence.	Documentation and later implementation should use authentic issues, commits, branches and Pull Requests.

Constraint Interactions and Trade-Offs

Scope vs Schedule:

An increase in scope brings additional requirements, implementation, testing, security and documentation work. Therefore, because the academic delivery period is fixed, uncontrolled scope growth could reduce the time available for verification and refinement.

This is why a scope statement is established prior to lower-level requirements and implementation decisions. PMI guidance points out that the definition of scope boundaries helps the team assess what requirements belong to the project and avoid uncontrolled expansion (Burek, 2008).

Engineering response:

Optional functionality should be deferred where it would jeopardize delivery of the core CivicConnect capabilities.

Resources vs Scope:

The amount of functionality that can be delivered depends on the development time, software tools, computing resources and external services available to the project. Adding unnecessary functionality may increase development, testing, documentation and deployment effort while also creating additional resource requirements.

Engineering response:

CivicConnect will prioritise its required business capabilities and favour technologies and services that are accessible, maintainable and appropriate for the available project resources.

Cost vs Deployment:

A free or low-cost hosting platform may reduce operational cost but may also impose restrictions relating to storage, performance, runtime support, availability or service usage.

Engineering response:

No deployment platform should be selected solely because it has a free tier. Deployment options should be compared and then selected based what is most suitable.

Security vs Usability:

Users need convenient services, but convenience is insufficient grounds for providing access to unnecessary information or functions.

Engineering response:

Following the principle of least privilege is the solution that helps to minimize privileges and provide only those necessary for the current task (OWASP Foundation, n.d.).

Initial Risk Register

Risk Management Approach

The CivicConnect Risk Register will be kept up-to-date and reviewed as the project progresses in accordance with the Master Project Brief to ensure that the risks to the project are sufficiently detailed to manage, and will consider areas such as requirements, people/skills, technology, security, integration, deployment, cost, schedule, quality and dependencies.

This approach is consistent with ISO 31000, which treats risk management as a continuing process involving risk identification, analysis, evaluation, treatment, monitoring and communication rather than a once-off exercise (ISO, 2018).

A project risk register commonly records information such as the risk, owner, cause, status, probability, impact and mitigation actions so that risks can be actively managed rather than merely listed (Project Management Institute, n.d.).

Risk Rating Method

Each identified risk will be given:

- a **Probability (P)** score from 1 to 5; and
- an **Impact (I)** score from 1 to 5.

The initial project risk exposure will be calculated as:

Risk Exposure = Probability × Impact

We will then use that score to determine the priority based on this scale that the team decided:

Score	Priority
1–4	Low
5–9	Medium
10–15	High
16–25	Critical

Initial CivicConnect Risk Register

Risk ID	Risk Event and Cause	P	I	Score	Priority	Mitigation	Contingency	Owner	Status
RSK-001	Uncontrolled additional features may enter the project after baseline because stakeholder requests are accepted without formal impact analysis.	4	4	16	Critical	Maintain clear in/out/future scope and use formal change control after baseline.	Defer non-essential functionality or formally revise approved scope after impact analysis.	Jordan	Open
RSK-002	Sensitive service-request information may be disclosed to unauthorized users because access requirements are incomplete or implemented incorrectly.	3	5	15	High	Define access/security requirements early and apply least-privilege principles during later design.	Restrict affected functionality or information until the access weakness is corrected and verified.	Darrin	Open
RSK-003	A later technology or deployment choice may be incompatible with the team's environment, capabilities or available services.	3	4	12	High	Defer final selection to M2 and evaluate alternatives using compatibility, team capability, cost, security and deployment evidence.	Replace the unsuitable option before major implementation dependency develops.	Darrin	Open
RSK-004	Requirements may become inconsistent or lose traceability as the project evolves.	3	4	12	High	Use stable requirement IDs, clear acceptance criteria and an evolving RTM.	Conduct traceability review and correct all affected artefacts before approving the change.	Jordan	Open
RSK-005	Quality evidence may be weak later because M1 requirements or NFRs are vague and cannot be objectively tested.	3	4	12	High	Define measurable NFRs and testable acceptance criteria during M1.	Revise inadequately testable requirements through controlled change before verification.	Darrin	Open
RSK-006	Service-request data may be lost or corrupted if data integrity, backup and recovery are	2	5	10	High	Record data integrity/recovery as a lifecycle concern before	Use the approved recovery process and investigate the underlying cause if loss occurs.	Darrin	Open

	considered too late.					persistence decisions are made.			
RSK-007	A free or low-cost platform may introduce unexpected operational limits or costs.	3	3	9	Medium	Investigate service limits and likely operational costs before committing to deployment technology.	Select an approved alternative or adjust deployment/resource usage if necessary.	Jordan	Open
RSK-008	Incorrect, incomplete or duplicate service-request data may be recorded because input validation and duplicate-handling rules are insufficient.	3	4	12	High	Define clear validation requirements and acceptance criteria; identify mandatory fields and duplicate-handling rules during requirements/design.	Correct affected records, strengthen validation rules and add regression tests for the failure cases.	Jordan	Open
RSK-009	Deployment may fail or behave differently from development because configuration, dependencies or environment settings are inconsistent between development, testing, staging and production.	3	4	12	High	Define environment requirements early, document configuration/dependencies and validate releases in a production-like staging environment.	Roll back to the last stable release, correct the configuration mismatch and redeploy after verification.	Darrin	Open
RSK-010	System failures or degraded performance may not be detected quickly because logging, monitoring or health-check mechanisms are inadequate.	2	4	8	Medium	Include observability requirements and plan appropriate logging, monitoring and health checks before deployment.	Investigate using available logs, restore service where possible and improve monitoring for the identified failure condition.	Jordan	Open

Highest-Priority Risk Analysis

At the initial baseline, RSK-001 — Scope Creep and RSK-002 Sensitive service-request — have the highest calculated exposure of 16/25 and 15/25.

RSK-001 — Scope Creep:

This risk is critical as each added feature introduces a range of additional activities beyond coding. Scope creep has an impact on requirements, design, UI, data, security, testing, documentation, schedule, and deployment/maintenance.

PMI literature flags uncontrolled scope growth as a risk and suggests a need to establish agreed mechanisms to control additions rather than tolerate them (Abramovici, 2000).

Risk response:

The team will:

- maintain a controlled scope baseline;
- clearly separate committed and future scope;
- evaluate proposed changes after baseline;
- consider their schedule/resource effect; and
- defer functionality where its cost outweighs the value achievable within the project constraints.

RSK-002 Sensitive service-request

RSK-004 has been given a high priority, since CivicConnect may contain information about service requests that should only be accessible by authorized users. If the requirements for access are incomplete or the access controls are implemented incorrectly, users might be able to view or change information that they are not allowed to see.

OWASP recommends applying the principle of least privilege, meaning that users should only be given the minimum permissions required to perform their responsibilities. OWASP also recommends designing authorisation controls early, denying access by default where appropriate, and verifying permissions on requests rather than relying only on what is displayed in the user interface (OWASP Foundation, n.d.).

Risk response:

- defining clear security and authorisation requirements during requirements engineering;
- identifying which user types should be permitted to perform specific actions;
- applying least-privilege principles when access-control rules are designed;
- considering sensitive information when defining data and interface requirements;
- ensuring later access-control implementation is tested through appropriate unit, integration and security tests; and
- recording any remaining security weaknesses as residual risks rather than claiming that the system is completely secure.

Functional Requirements

These describe the specific features the system must perform.

ID	Priority	Description	Source
FR-01	Must-have	The system shall allow citizens to create a new account using an email address and password.	Citizen Stakeholder
FR-02	Must-have	The system shall allow citizens to submit a new service request, including title, description, issue category (e.g., Roads, Water, Garbage), and location.	Citizen Stakeholder
FR-03	Should-have	The system shall allow citizens to attach photos or videos to their service request as supporting evidence.	Citizen Stakeholder
FR-04	Must-have	The system shall allow citizens to view the current status of their submitted requests.	Citizen Stakeholder
FR-05	Must-have	The system shall allow municipal staff (admins) to view a complete list of all submitted requests.	Municipal Authority Stakeholder
FR-06	Must-have	The system shall allow admins to filter and search requests by category, status, and date.	Municipal Authority Stakeholder
FR-07	Must-have	The system shall allow admins to update the status of a request (e.g., from "Pending" to "In Progress" to "Resolved").	Municipal Authority Stakeholder
FR-08	Should-have	The system shall allow admins to add official replies to a citizen's request thread.	Municipal Authority Stakeholder
FR-09	Could-have	The system shall allow citizens to upvote other citizens' issues to highlight community priorities.	Community Engagement Goal
FR-10	Won't-have (Future)	The system shall be integrated with an external payment gateway for permit applications.	Deferred Scope

Non-Functional Requirements

These define the qualities and constraints of the system.

ID	Category	Description	Source
NFR-PERF-01	Quality/Performance	The citizen dashboard page shall load in under 2 seconds for 95% of requests.	Quality Constraint
NFR-SEC-01	Security	All user passwords shall be hashed and salted before storage.	Security Constraint
NFR-SEC-02	Security	All data transmitted between the client and server shall be encrypted using TLS 1.2 ¹ or higher.	Security Constraint
NFR-REL-01	Quality/Reliability	The system shall maintain an uptime of 99.5% during business hours.	Quality Constraint
NFR-COST-01	Cost/Resource	The development and operational costs must remain within the allocated budget of R500/month.	Cost Constraint
NFR-SCALE-01	Quality/Scalability	The system shall support up to 5,000 concurrent users without significant degradation in performance.	Quality Constraint
NFR-COMP-01	Compliance	The system shall comply with the POPIA regulations for data privacy and protection.	Scope/Quality Constraint

NFR-OBS-01	Observability	The system shall log all CRUD operations on requests for auditing purposes.	Quality Constraint
------------	---------------	---	--------------------

Acceptance Criteria

Clear, testable conditions for a requirement to be considered "done." (Cucumber, 2026)

For FR-02: Submit a Service Request

- **Given** that a citizen is logged into the system
- **When** they navigate to the "Submit Request" form, fill in a valid title, description, and category, and click "Submit"
- **Then** the system shall create a new service request with a unique ID, status "Pending," and display a confirmation message to the user. (Cucumber, 2026)

For FR-07: Admin Updates Request Status

- **Given** that an admin is logged into the system
- **When** they select a pending request from the list, change its status to "In Progress," and save the change
- **Then** the system should update the request's status and timestamp in the database, and the change shall be immediately reflected in the citizen's view. (Cucumber, 2026)

For NFR-PERF-01: Performance

- **Given** a simulated load of 1,000 concurrent users
- **When** they request the citizen dashboard
- **Then** the page load time for 95% of requests must be under 2 seconds. (ISO/IEC, 2023)

For NFR-SEC-01: Security

- **Given** a new user registration or a password change
- **When** the user submits their password
- **Then** the system must store it as a hashed value using a strong algorithm (e.g., bcrypt) and not as plain text. (OWASP Foundation, 2026)

Initial RTM

Requirement ID	Requirement Description	Source	Priority	Acceptance Criteria ID
FR-01	User account creation	Citizen Stakeholder	Must-have	AC-FR-01
FR-02	Submit service request	Citizen Stakeholder	Must-have	AC-FR-02
FR-07	Admin updates request status	Municipal Authority	Must-have	AC-FR-07
NFR-PERF-01	Dashboard load under 2 seconds	Quality Constraint	Should-have	AC-NFR-PERF-01
NFR-SEC-01	Passwords hashed	Security Constraint	Must-have	AC-NFR-SEC-01

NFR-COMP-01	POPIA compliance	Scope/Quality Constraint	Could-have	AC-NFR-COMP-01
-------------	------------------	--------------------------	------------	----------------

Engineering Decision Log

The Engineering Decision Log is used to capture the rationale behind key decisions made during the development of CivicConnect. It will serve as a living document that explains the reasoning behind major decisions and ensures that important decisions are justified based on the information and limitations available at the time the decision was made.

Decision Status

The following statuses will be used:

Status:	Meaning:
Accepted	The decision has been agreed and currently applies to the project.
Deferred	A decision is required later, but there is currently insufficient evidence to make it responsibly.
Superseded	A later approved decision has replaced the original decision.
Rejected	An alternative was considered but deliberately not selected.

Initial Engineering Decision Register

Decision ID	Decision	Status
DEC-001	Priorities the mandatory CivicConnect business capabilities as the initial committed scope.	Accepted
DEC-002	Defer direct WhatsApp integration from the initial project scope.	Accepted
DEC-003	Use a 5 × 5 probability-impact method to priorities initial project risks.	Accepted
DEC-004	Defer final application delivery approach until M2.	Deferred
DEC-005	Defer final technology stack and software architecture until M2.	Deferred
DEC-006	Establish the need for controlled role-based access now, but defer the exact authentication and authorization mechanism.	Deferred
DEC-007	Defer the final deployment/hosting platform until appropriate alternatives have been evaluated.	Deferred

Forward Engineering Considerations

1. Security (OWASP² Top 10)

- **Why it matters now:** Security requirements (NFR-SEC-01, NFR-SEC-02) must be defined now to guide M2 architecture and implementation. (OWASP Foundation, 2021)
- **Risk of ignoring:** Introduction of vulnerabilities (e.g., SQL injection, broken authentication), leading to data breaches and non-compliance with POPIA. (OWASP Foundation, 2021)

2. Scalability and Performance

- **Why it matters now:** NFR-PERF-01 and NFR-SCALE-01 define performance expectations. These influence database schema design and architectural choices (e.g., caching, load balancing). (ISO/IEC, 2023)
- **Risk of ignoring:** System becomes slow during peak usage, failing NFRs and leading to user dissatisfaction. (ISO/IEC, 2023)

3. Testability (Testing Strategy)

- **Why it matters now:** Requirements must be testable. Acceptance criteria are the first step. M2 architecture must support unit, integration, and performance testing.
- **Risk of ignoring:** Late defect discovery, making fixes expensive and causing unreliable releases.

4. Observability (Logging & Monitoring)

- **Why it matters now:** NFR-OBS-01 requires logging. This must be considered in architecture to ensure all CRUD operations are auditable.
- **Risk of ignoring:** Inability to debug issues in production, identify security breaches, or measure system performance against SLAs. (OWASP Foundation, 2021)

5. Deployment Environment

- **Why it matters now:** The target environment (cloud vs. on premise) impacts cost, security, and technology choices.

Risk of ignoring: Choosing a technology stack or architecture that is not cost-effective or is difficult to deploy in the target environment.

6. Maintainability

- **Why it matters now:** The decision to use a 3-tier architecture (D-01) was made to ensure maintainability. (Microsoft, 2021).
- **Risk of ignoring:** High technical debt making the system hard to modify in the future, increasing operational costs.

7. Responsible Use of AI (as per Assignment 1)

- **Why it matters now:** Generative AI tools could be used to help draft requirements, code, or tests.
- **Risk of ignoring:** Introducing insecure or incorrect code that appears correct, or creating hidden technical debt. Aligns with OWASP Top 10 for LLM applications. (OWASP Foundation, 2026).

PED v1.0 Baseline Review and Sign-Off

Item	Details
Project	CivicConnect
Baseline Type	Milestone 1 — Engineering Foundation & Requirements Baseline
Version	PED v1.0
Date	2026/09/09
Scope reviewed	YES
Requirements and traceability checked	YES
Risk review completed	YES
Repository/governance controls checked	YES
Outcome	ACCEPTED

Team Approval

Team Member	Review Confirmation	Date
Jordan Naidoo	Approved	2026/09/09
Darrin Everette Mokuena	Approved	2026/09/09

AI Verification:

Team Member: Jordan Naidoo

AI tool / use: Co Pilot

Purpose: Risk register structure and calculation assistance, risk analysis format

Output used: Calculations for risk register

What was changed/rejected: parts of risk analysis format

Team Member: Darrin Everette Mokuena

AI tool / use: Co Pilot

Purpose: what is needed in an Acceptance criterion

Output used: general format

What was changed/rejected: parts of format were changed

References

Abramovici, A. 2000. Controlling Scope Creep. PMNetwork, 14(1), pp. 44–48. Project Management Institute. PMI – Controlling Scope Creep

Belgium Campus ITversity. 2026a. SEN381 Software Engineering 381: Integrated Team Software Engineering Project — CivicConnect Master Project Brief. Academic Year 2026.

Belgium Campus ITversity. 2026b. SEN381 Software Engineering 381: CivicConnect Project — Milestone 1: Engineering Foundation & Requirements Baseline. Academic Year 2026.

Burek, P. 2008. Creating Clear Project Requirements: Differentiating “What” from “How”. Paper presented at PMI Global Congress 2008 — North America, Denver, Colorado. Newtown Square, PA: Project Management Institute.

Cucumber. n.d. Gherkin Reference. Cucumber Documentation. Accessed 9 September 2026.

Faris, R.K. and Patterson, D. 2007. Managing Risk in the Project Portfolio. Paper presented at PMI Global Congress 2007 — North America, Atlanta, Georgia. Newtown Square, PA: Project Management Institute.

International Organization for Standardization (ISO). 2018. ISO 31000:2018 Risk Management — Guidelines. Geneva: International Organization for Standardization.

ISO/IEC. 2023. ISO/IEC 25010:2023 Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE) — Product Quality Model. Geneva: International Organization for Standardization.

ISO/IEC/IEEE. 2018. ISO/IEC/IEEE 29148:2018 Systems and Software Engineering — Life Cycle Processes — Requirements Engineering. Geneva: International Organization for Standardization.

Microsoft. 2021. Using a Three-Tier Architecture Model. Microsoft Learn. Accessed 9 September 2026.

OWASP Foundation. 2021. OWASP Top 10:2021. Open Worldwide Application Security Project.

OWASP Foundation. n.d. Access Control. Open Worldwide Application Security Project. Accessed 9 September 2026.

OWASP Foundation. n.d. Password Storage Cheat Sheet. OWASP Cheat Sheet Series. Accessed 9 September 2026.

OWASP GenAI Security Project. 2026. OWASP GenAI LLM Top 10 2026. Open Worldwide Application Security Project.

Republic of South Africa. 2013. Protection of Personal Information Act 4 of 2013. Pretoria: Government of South Africa.

Systems Engineering Body of Knowledge (SEBoK). 2026. Stakeholder Needs and Requirements. SEBoK Version 2.14.